GRADE 6 MATHEMATICS · REVEAL MATH

# Unit 3

# Ratios and Rates

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**Lesson 3-1** Understanding Ratios · Ratios and Tape Diagrams

**Lesson 3-2** Understanding Unit Rates · Comparing Unit Prices (Best Buy)

**Lesson 3-3** Equivalent Ratios and Tables · The Bridge Strategy (Scaling Ratios)

**Lesson 3-4** Graphing Ratios on the Coordinate Plane · Interpreting Points on Ratio Graphs

**Lesson 3-5** Comparing Ratios with Common Terms · Comparing Rates with Benchmarks

**Lesson 3-6** Customary Unit Conversions · Metric Unit Conversions

**Lesson 3-7** Customary vs. Metric Conversions · Multi-Step System Conversions and Dosage

# What's in each lesson folder

| | |
|---|---|
| **N.M.pptx** | Teaching slides for both sessions (warm-up, launch, worked examples, practice). |
| **Warm-Up** | One page of skills the lesson assumes. Answers: Unit Warm-Up Key. |
| **Prerequisite Practice** | A and B sheets per session for students who need a missing skill first, plus a key. |
| **Guided Notes** | Core notes plus model/reasoning pages for both sessions. Select the current session pages in Print Packets. |
| **Practice** | Form A (class) and Form B (reteach / retake) per session. 11 numbered problems with subparts. Use the lesson route to select the core set; see the current PDF for page count. |
| **Exit Tickets** | One page per session, two tickets per page. Three quick items and a self-rating. |
| **Homework** | Core: 1 page per session. Supported: 2 pages per session; print duplex as 2 sheets. Same problems and answers, with language supports. |
| **Answer Key** | Every answer for notes, practice (both forms), exit tickets and homework; talk prompts, partner task, misconceptions; item analysis. |

## A lesson in two sessions

| | |
|---|---|
| **Session 1** | Warm-Up → slides + Guided Notes (page 1) → Practice Form A → Exit Ticket → Homework Sheet 1 |
| **Session 2** | Slides + Guided Notes (page 2) → Practice Form A → Exit Ticket → Homework Sheet 2 |
| **Next day** | Use the lesson Teaching Support Guide to match errors to feedback, revision and a fresh check. Confidence ratings inform support; they do not replace work evidence. |

# Lesson map

| Lesson | Session | Standard | Learning target |
|---|---|---|---|
| **3-1** | 1 · Understanding Ratios | 6.RP.A.1 | I can describe a relationship between two quantities using ratio language. |
| **3-1** | 2 · Ratios and Tape Diagrams | 6.RP.A.1 | I can use a tape diagram to share a total according to a given ratio. |
| **3-2** | 1 · Understanding Unit Rates | 6.RP.A.2 | I can find a unit rate and use rate language such as “per hour” or “each.” |
| **3-2** | 2 · Comparing Unit Prices (Best Buy) | 6.RP.A.2 | I can compare unit prices to decide which package is the better buy. |
| **3-3** | 1 · Equivalent Ratios and Tables | 6.RP.A.3a | I can build a ratio table by scaling both rows by the same factor. |
| **3-3** | 2 · The Bridge Strategy (Scaling Ratios) | 6.RP.A.3a | I can use the bridge strategy to scale a ratio when I cannot multiply by a whole number. |
| **3-4** | 1 · Graphing Ratios on the Coordinate Plane | 6.RP.A.3a | I can turn a ratio into ordered pairs and plot them on a coordinate plane. |
| **3-4** | 2 · Interpreting Points on Ratio Graphs | 6.RP.A.3a | I can explain what a point on a ratio graph means in the situation it describes. |
| **3-5** | 1 · Comparing Ratios with Common Terms | 6.RP.A.3b | I can compare two ratios by scaling them until they share a common term. |
| **3-5** | 2 · Comparing Rates with Benchmarks | 6.RP.A.3b | I can use a benchmark to compare two rates without dividing. |
| **3-6** | 1 · Customary Unit Conversions | 6.RP.A.3d | I can convert between customary units by multiplying or dividing by the conversion rate. |
| **3-6** | 2 · Metric Unit Conversions | 6.RP.A.3d | I can convert between metric units by multiplying or dividing by powers of ten. |
| **3-7** | 1 · Customary vs. Metric Conversions | 6.RP.A.3d | I can convert a measurement between the customary and metric systems using a given rate. |
| **3-7** | 2 · Multi-Step System Conversions and Dosage | 6.RP.A.3d | I can solve a multi-step problem that requires converting units before using a rate. |

# Teaching, assessment, and retrieval route

Use the current lesson Teaching Support Guide for copy counts, planned 45-minute route, item-specific feedback, and a reserved recheck. Student notes now include full-size conceptual task pages; select them deliberately rather than assigning every page in one session.

| Resource | Use | Key/response |
|---|---|---|
| Readiness Check | Before the unit; identify access to upcoming skills. | Matching Teacher Key; plan prerequisites without labeling students. |
| Midunit Check | After the first half of the unit. | Review the item scope before assigning; use the matching key. |
| End Form A | After all relevant targets have been taught. | Separate Teacher Key; item criteria and partial credit. |
| End Form B | After revision, only if unseen. | Parallel Teacher Key; independent evidence. |
| Synthesis Task | Connect several taught representations and decisions. | Worked anchor and an 8-point analytic rubric. |

Readiness and midunit checks are short teacher-authored samples, not exhaustive tests. The end forms also sample the unit; use lesson exit evidence for targets not sampled. Preserve the existing Unit 3 Version B test as a separate publisher-aligned retained assessment.

## Retrieval and intervention

3.1: retrieve one independent item after two lessons and about one week; use the lesson guide to target model, arithmetic, or explanation errors.

3.2: retrieve one independent item after two lessons and about one week; use the lesson guide to target model, arithmetic, or explanation errors.

3.3: retrieve one independent item after two lessons and about one week; use the lesson guide to target model, arithmetic, or explanation errors.

3.4: retrieve one independent item after two lessons and about one week; use the lesson guide to target model, arithmetic, or explanation errors.

3.5: retrieve one independent item after two lessons and about one week; use the lesson guide to target model, arithmetic, or explanation errors.

3.6: retrieve one independent item after two lessons and about one week; use the lesson guide to target model, arithmetic, or explanation errors.

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